

Exploring a 2×2 Factorial Design With Control and Blocks: A Case Study in R

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Introduction

Factorial experiments are widely used in scientific research for evaluating the effects of multiple factors simultaneously. The simplest form of the factorial design is a 2×2 factorial design which involves two factors. For each factor we have two levels, resulting in four treatment combinations. However, in many practical scenarios, researchers may wish to compare these treatments not only among themselves but also against a standard or baseline treatment referred to as a Control. In such cases, a 2×2 factorial design with a Control is adopted. There is an important difference between the traditional factorial design with a control and the problem which we attempt to address in this chapter. With the traditional factorial design with a control, we see all possible combinations of the factor levels including the Control. For example, we may consider an experiment with two factors, type of fertilizer and irrigation each with two levels respectively. If A and B are the levels of fertilizer types and the levels of irrigation are Yes and No, where the latter is a control. Then for a traditional 2×2 factorial design with control, we would have all possible combinations: (A, Yes), (A, No), (B, Yes) and (B, No). Nevertheless, the type of problems that we try to address in this chapter involves a 2×2 factorial design where the control is serves as the fifth treatment. We begin this chapter with a detailed discussion on a 2×2 factorial design with control as the fifth treatment and extend our discussion to a 2×2 factorial design with control and blocks for a better understanding of our readers.

In this chapter, we illustrate the concept of a 2×2 factorial design with control, where the control is considered as the fifth treatment using the example below. Researchers are interested in evaluating the effects of two drugs on blood pressure. They would like to determine if a Soluble or Non-soluble form makes a difference. Another factor that affects performance of the drug is dosage (10 units and 15 units). Because this is a new study, they also chose a Control group to serve as reference. In this experiment, the Control group refers to experimental units that did not receive any of the treatments.

A 2×2 factorial design with control may also be utilized in experiments where methods are being tested for the first time or in new areas of study. In the case where methods are being tested for the first time, researchers may be interested in comparing the main effects of treatments with Control. In the example we consider, we are interested in determining if the treatments are able to reduce blood pressure at all. In such instances, the control group serves as a baseline and enables us to answer such a question. That is, if we find out that the blood pressure of individuals taking any of the drugs at different concentrations are on average lower than those in the control group, then we know that taking a drug is at least better than taking nothing at all. Also, in situations where there may be a significant interaction effect, we can compare each treatment combination with the control.

Statistical Challenge

As we stated in the introduction, with a 2×2 factorial design with a control, the control is considered as the fifth treatment and does not appear in any other treatment combination unlike the traditional 2×2 factorial design with Control.

The statistical challenge is that, the control is not part of the factorial structure and thus, it creates a non-orthogonal or partially unbalanced design. This is a problem because the control does not contribute to the estimation of the main effects and interaction effects of factors. Nevertheless, the control shares the same experimental error term with the treatments. When the traditional 2×2 factorial design is used to analyze these kinds of experiments, the implication is that the control contributes to the marginal means and also, both main effects and interactions are interpretable and balanced. This is incorrect.

The 2×2 factorial design with control enables the mean of the control to be estimated separately and computes factorial means only from the four factorial cells. Moreover, the variance of factorial effects are estimated only from factorial units and thus, control observations do not influence variance.

Exploratory Data Analysis

In our example which will be used to demonstrate the ideas in this chapter, we first conduct an exploratory data analysis to understand how data was generated, the type and description of the variables. We only present information that will be useful to the reader when using this design. We show the histogram which gives us information about the distribution of the response. We also show the interaction plot and boxplot for all treatment combinations and the control.

Recall, the objective of the study is to evaluate the effects of two drugs on blood pressure. They would like to determine if a Soluble or Non-soluble form makes a difference. Another factor that affects performance of the drug is dosage (10 units and 15 units).

Data Description

Data was simulated from the normal distribution with mean of 0 and variance of 5. To set up the 2×2 factorial design with control in R, we first set up the treatment combination of drug type and dosage as a traditional factorial design and only add the control as an additional factor.

There were 15 observations with three replicates from each treatment combination and the control. Both drug type (Soluble and Non-soluble) and dosage (10 units and 15 units) were considered as qualitative variables. Thus, the reader can think of the levels of dosage as low

and high for simplicity. The response variable in this experiment is blood pressure. Each individual in this experiment is considered as an experimental unit.

```
# Create the data frame
layout <- data.frame(
  Drug_Type = c("Soluble", "Soluble",
                "Non-soluble", "Non-soluble", "No Drug"),
  Dosage = c(10, 15, 10, 15, 0),
  Description = c(
    "Soluble at low dosage",
    "Soluble at high dosage",
    "Non-soluble at low dosage",
    "Non-soluble at high dosage",
    "Control"
  ),
  Replicates = rep(3, 5)
)

# Create table
kable(
  layout,
  caption = "Experimental Drug Layout",
  col.names = c("Drug Type", "Dosage", "Description", "Replicates")
)
```

Table 1: Experimental Drug Layout

Drug Type	Dosage	Description	Replicates
Soluble	10	Soluble at low dosage	3
Soluble	15	Soluble at high dosage	3
Non-soluble	10	Non-soluble at low dosage	3
Non-soluble	15	Non-soluble at high dosage	3
No Drug	0	Control	3

Shows the layout of the experiment. Drug Type has two levels (Soluble and Non-soluble) and Dosage has two levels (10 and 15 units) with control serving as a reference.

```

## Setting seed for reproducibility
set.seed(2025)

# Define levels for drug type
drug_type <- c("Soluble", "Non-soluble")
# Define levels for dosage
dosage <- c(10, 15)
# Define number of replicates
replicates <- 3

# Generate factorial combinations
factorial_df <- expand.grid(Drug = drug_type,
                           Dosage = dosage)
factorial_df <- factorial_df[rep(1:nrow(factorial_df),
                                each = replicates), ]

# Simulate blood pressure (response variable)
factorial_df$BloodPressure <- with(factorial_df,
                                   130 -
                                   ifelse(Drug == "Soluble", 5, 0) +
                                   ifelse(Dosage == 15, 3, 0) +
                                   ifelse(Drug == "Soluble"
                                          & Dosage == 15, -2, 0) +
                                   rnorm(nrow(factorial_df),
                                          mean = 0, sd = 5)
)

# Add Control group
control_df <- data.frame(
  Drug = "control",
  Dosage = 0,
  BloodPressure = rnorm(replicates, mean = 135, sd = 5)
)

# Put the two data frames together by row
bp_data <- rbind(factorial_df, control_df)

# Convert drug type and dosage to factors since they are qualitative variables
bp_data$Drug <- as.factor(bp_data$Drug)
bp_data$Dosage <- as.factor(bp_data$Dosage)

```

```

# Creating a new variable Control
# Control serves as an indicator variable for whether control is present or not

bp_data <- bp_data %>%
  mutate(Control = ifelse( Dosage== 0, "y", "n"))
bp_data$Control <- as.factor(bp_data$Control)

# Create formatted table
kable(
  bp_data,
  caption = "Simulated Blood Pressure Data",
  col.names = c("Drug Type", "Dosage",
                 "Blood Pressure", "Control")
)

```

Table 2: Simulated Blood Pressure Data

	Drug Type	Dosage	Blood Pressure	Control
1	Soluble	10	128.1038	n
1.1	Soluble	10	125.1782	n
1.2	Soluble	10	128.8658	n
2	Non-soluble	10	136.3624	n
2.1	Non-soluble	10	131.8549	n
2.2	Non-soluble	10	129.1857	n
3	Soluble	15	127.9856	n
3.1	Soluble	15	125.6001	n
3.2	Soluble	15	124.2752	n
4	Non-soluble	15	136.5108	n
4.1	Non-soluble	15	131.0215	n
4.2	Non-soluble	15	124.2247	n
11	control	0	132.8952	y
21	control	0	138.8245	y
31	control	0	140.3308	y

Shows the simulated data which has three replicates for each treatment combination.

Figure 1 shows the relative frequency density plot which displays the distribution of blood pressure measurements in the data. The x-axis represents blood pressure values, while the y-axis shows relative frequency density, meaning the total area under the bars sums to one. Most observations are concentrated between approximately 124 and 136, indicating this range contains the highest relative density of measurements. The distribution shows moderate spread,

with fewer observations at the higher end near 140 mmHg, suggesting slight right-skewness. No extreme outliers are apparent, although the density decreases toward the upper tail. Overall, the plot suggests that blood pressure values are clustered around the mid-120s to mid-130s, with relatively low variability outside this range.

```
## Histogram plot

ggplot(bp_data, aes(x = BloodPressure)) +
  geom_histogram(
    bins = 10,
    color = "black",
    fill = "gray"
  ) +
  labs(
    title = "Histogram showing the distribution of blood pressure",
    x = "Blood Pressure",
    y = "Frequency"
  ) +
  theme_minimal()
```

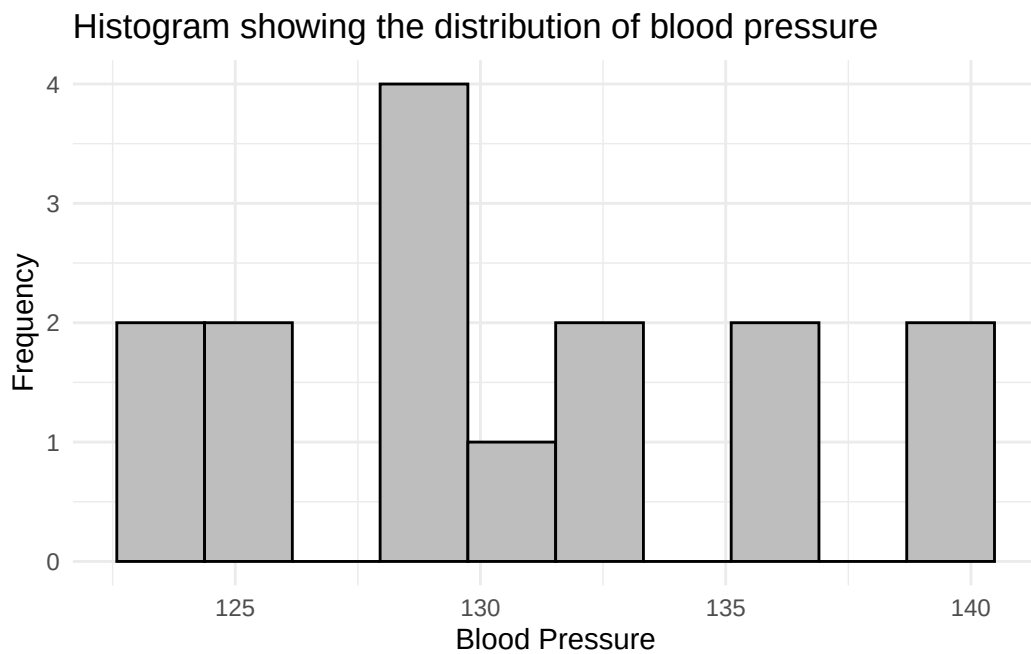


Figure 1: Distribution of Blood Pressure

Figure 2 displays the distribution of blood pressure across different drug types (Soluble, Non-soluble, and Control) at three dosage levels (0, 10, and 15 units). At dosage 0, the control group exhibits the highest median blood pressure, with values centered around the upper 130s, and relatively low variability. This suggests elevated blood pressure levels in the absence of treatment. No soluble or non-soluble drug groups are present at this dosage level.

At dosage 10, both soluble and non-soluble drugs show reduced median blood pressure compared to the control at dosage 0. The soluble drug group has the lowest median blood pressure and a relatively narrow interquartile range, indicating more consistent responses. In contrast, the non-soluble drug group shows a higher median and greater variability, suggesting less uniform treatment effects.

At dosage 15, the soluble drug continues to demonstrate the lowest median blood pressure with minimal variability, indicating a strong and consistent treatment effect. The non-soluble drug group again shows higher median values and a wider spread, with observations extending across a broader range, reflecting increased variability in response.

```
# Create boxplots of blood pressure for each drug type
ggplot(bp_data, aes(x = Drug, y = BloodPressure, fill = Drug)) +

  # Draw boxplots
  geom_boxplot() +

  # Create separate panels for each dosage level
  facet_wrap(~ Dosage) +

  # Add plot title and axis labels
  labs(
    title = "Boxplot of Blood Pressure by Dosage and Drug Type",
    x = "Drug",
    y = "Blood Pressure"
  ) +

  # Apply a clean theme
  theme_minimal() +

  # Rotate x-axis labels for better readability
  theme(
    axis.text.x = element_text(
      angle = 90,
      vjust = 0.5,
      hjust = 1
    )
  )
```

)

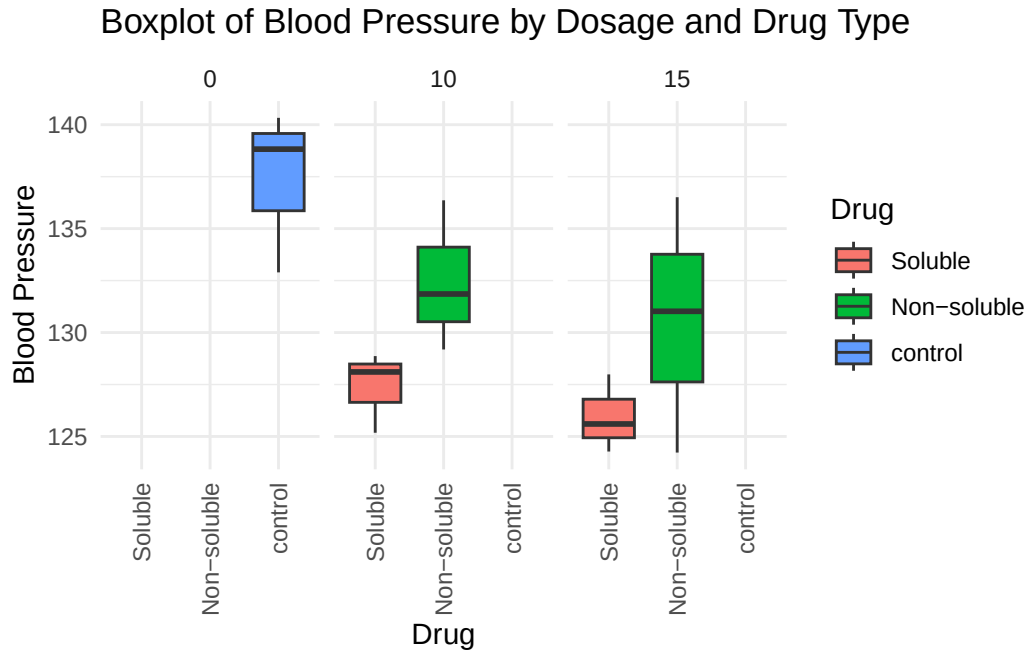


Figure 2: Distribution of BloodPressure by Drug Type and Dosage

Figure 3 shows the interaction plot for drug type and dosage. The absence of an interaction for a 2×2 factorial design is usually illustrated by parallel lines. From the interaction plot shown in Figure 3, we see parallel lines which shows there is no interaction between the two factors.

This implies that the effect of one factor, say drug type on blood pressure, is the same at all levels of dosage. Thus, the two factors are independent of each other. Nevertheless, we need to look at the ANOVA table.

```

# Remove the control group so only factorial treatments are used
bp_data_sub <- subset(bp_data, Drug != "Control")

# Create interaction plot
interaction.plot(

  # Factor shown on the x-axis
  x.factor = bp_data_sub$Drug,

  # Factor represented by separate lines
  trace.factor = bp_data_sub$Dosage,

  # Response variable
  response = bp_data_sub$BloodPressure,

  # Plot points connected by lines
  type = "b",

  # Line colors for dosage levels
  col = c("blue", "red"),

  # Use different line types:
  # 2 = dashed line for dosage 10
  # 1 = solid line for dosage 15
  lty = c(2, 1),

  # Line width
  lwd = 2,

  # Plotting symbol
  pch = 16,

  # Axis labels and title
  ylab = "Average Blood Pressure",
  xlab = "Drug",
  trace.label = "Dosage",
  main = "Interaction Plot of Drug and Dosage \n (with Control Reference)"
)

# Add horizontal line showing control group mean
abline(

```

```

h = mean(bp_data$BloodPressure[bp_data$Drug == "Control"]),
col = "darkgreen",
lty = 2,
lwd = 2
)

# Add legend for control mean line
legend(
  "topleft",
  legend = "Control Mean",
  col = "darkgreen",
  lty = 2,
  lwd = 2,
  bty = "n"
)

```

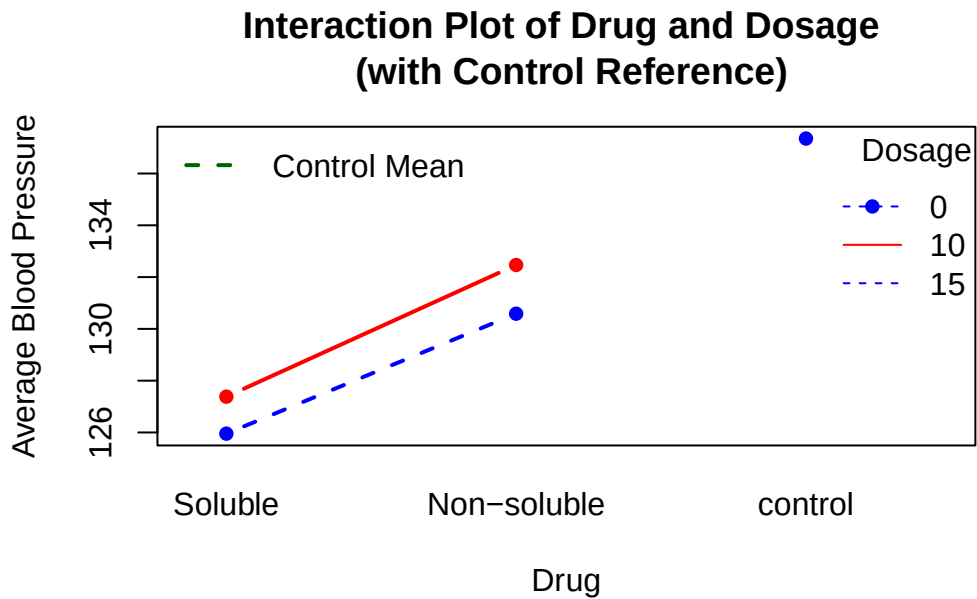


Figure 3: Interaction Plot of Drug Type and Dosage with Control as reference (shown in green color)

Advanced Analysis

Recall, the main objective of the experiment is to understand how drug type and dosage levels affect blood pressure and also how the treatments perform in comparison to the control. Thus, we seek to answer two important questions with a single model.

The statistical model thus seek to evaluate whether average blood pressure of individuals taking a soluble drug differs from the blood pressure of individuals with non-soluble drug. Moreover, the model enables us to compare average blood pressure when a treatment was administered to that of the control group who did not receive any treatment all.

Similarly, we are able to evaluate the effect of the dosage levels and compare their effect on blood pressure. In this chapter, we consider a simple linear model with assumption that blood pressure is independent and identically distributed and that we have homogeneous variance.

Statistical Model

The linear model for a factorial design with control is given as;

$$Y_{ij} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \gamma I_k + \varepsilon_{ijk}$$

and,

$$I_k = \begin{cases} 1, & \text{if the observation belongs to the control treatment,} \\ 0, & \text{if the observation belongs to one of the factorial treatments.} \end{cases}$$

$i = 1, 2, j = 1, 2, 3$ and $k = 0, 1$

where:

y_{ijk} is the blood pressure of the l th individual using the i th type of drug at the j th dose

μ represents the baseline mean of the factorial treatments

α_i is the effects of the i th drug type

β_j is the effects of the j th dosage

$(\alpha\beta)_{ij}$ is their interaction effect

γ measures the difference between the control treatment and the factorial baseline

ε_{ijk} is assumed to be independently and normally distributed with mean zero and constant variance σ^2 .

Table 3 shows the ANOVA table when the data is analyzed as a traditional 2 by 2 factorial design. Note that, analyzing the experiment in this manner is incorrect. Looking at the Figure 4, we see that there is a significant effect of dosage with $p\text{-value} = 0.02$. This is because the control is embedded in dosage and thus the estimated mean for dosage includes that of control. With this result, a researcher may believe that dosage of the drug was significant and may continue to explore which dosage led to the highest effect.

```
library(lme4)
# Analyzing as a traditional 2 by 2 factorial
bp.m0 <- lm(BloodPressure ~ Dosage + Drug + Drug*Dosage, data = bp_data)

#summary(bp.m0)
anova_tab <- anova(bp.m0)

# Print as formatted table
kable(
  anova_tab,
  caption = "ANOVA Table for 2x2 Factorial Model"
)
```

Table 3: ANOVA Table for 2x2 Factorial Model

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Dosage	2	171.6828239	85.8414119	5.8146609	0.0211245
Drug	1	70.8175226	70.8175226	4.7969840	0.0533177
Dosage:Drug	1	0.1539226	0.1539226	0.0104263	0.9206885
Residuals	10	147.6292669	14.7629267	NA	NA

ANOVA table when data is analyzed as a traditional 2 by 2 factorial design

Table 4 takes a look at the correct model. From Table 4, we see that the $p\text{-value}$ for the control shown in the ANOVA table is statistically significant, $p\text{-value} = 0.01$ and the $p\text{-value}$ for dosage by drug interaction term is 0.92. These values correspond to the interaction plot we saw earlier. Since there is no significant interaction effect, we can look at the main effects for Drug (soluble or non-soluble) to determine if there is any significant difference.

```

# Analyzing as a traditional 2 by 2 factorial with Control
bp.m1 <- lm(BloodPressure ~ Control + Dosage + Drug + Drug*Dosage, data = bp_data)

#summary(bp.m1)
anova_tab1 <- anova(bp.m1)

# Print as formatted table
kable(
  anova_tab1,
  caption = "ANOVA Table for 2x2 Factorial Model"
)

```

Table 4: ANOVA Table for 2x2 Factorial Model

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Control	1	163.4607459	163.4607459	11.0723808	0.0076498
Dosage	1	8.2220780	8.2220780	0.5569409	0.4726749
Drug	1	70.8175226	70.8175226	4.7969840	0.0533177
Dosage:Drug	1	0.1539226	0.1539226	0.0104263	0.9206885
Residuals	10	147.6292669	14.7629267	NA	NA

ANOVA table when data is analyzed using 2 by 2 factorial design with control

We proceed with further analysis by looking at the main and simple effects of only the correct model. Table 5 shows the estimated marginal means for drug type. For drug type soluble and non-soluble, the estimated marginal means are 127 and 132 respectively with standard error 1.57. The estimated mean for the control group is also 137 with standard deviation of 2.22. The 95 percent confidence intervals are also shown in the last two columns of the same table.

```
# Estimated marginal means for drug type incorrect model
bp.m1.emmeans<- emmeans (bp.m1, ~ Drug)

# Print as formatted table
kable(
  bp.m1.emmeans,
  caption = "Estimated marginal means of drug type(Incorrect Model)"
)
```

Table 5: Estimated marginal means of drug type(Incorrect Model)

	Drug	Control	emmean	SE	df	lower.CL	upper.CL
1	Soluble	n	126.6681	1.568594	10	123.1730	130.1631
2	Non-soluble	n	131.5267	1.568594	10	128.0316	135.0217
6	control	y	137.3502	2.218327	10	132.4074	142.2929

showing the estimated marginal means of drug type when data is analyzed as a 2 by 2 factorial design with a control

Table 6 shows the estimated marginal means for dosage. When dosage levels are at 10 and 15 units, the estimated marginal means are 130 and 128 respectively with standard error 1.57. The estimated mean for the control group is again 137 with standard deviation of 2.22 indicating that the control group did not receive any treatment.

```
# Estimated marginal means for drug type incorrect model
bp.m2.emmeans<- emmeans (bp.m1, ~ Dosage)

# Print as formatted table
kable(
  bp.m2.emmeans,
  caption = "Estimated marginal means of drug type(Correct Model)"
)
```


Table 6: Estimated marginal means of drug type(Correct Model)

	Dosage	Control	emmean	SE	df	lower.CL	upper.CL
2	10	n	129.9251	1.568594	10	126.4301	133.4202
3	15	n	128.2696	1.568594	10	124.7746	131.7647
4	0	y	137.3502	2.218327	10	132.4074	142.2929

showing the estimated marginal means of dosage when data is analyzed as a 2 by 2 factorial design with a control

Simple Effects

Next, we show the simple effects of drug and dosage. A simple effect refers to the effect of one treatment factor at a specific level of another factor. One typically would need to look at simple effects if there is a significant interaction effect. It is used in factorial experiments to understand how factors interact. In our example, there was no significant interaction effect but we show the results here for completeness.

From Table 7, the simple effect for dosage at 10 and 15 units when a soluble drug is used are 127 and 126 respectively. For non-soluble drug, the simple effect of dosage at 10 and 15 units are 132 and 131 respectively. Looking at the simple effects, we see a lower blood pressure when the drug is soluble. Nevertheless, without any drug or dosage, the blood pressure is high.

```
# Simple effects for dosage at each drug type level
bp.m1.emmeans2 <- emmeans (bp.m1, ~ Dosage | Drug)

# Print as formatted table
kable(
  bp.m1.emmeans2,
  caption = "Simple effects of dosage at the given drug types"
)
```

Table 7: Simple effects of dosage at the given drug types

	Dosage	Drug	Control	emmean	SE	df	lower.CL	upper.CL
2	10	Soluble	n	127.3826	2.218327	10	122.4398	132.3253
3	15	Soluble	n	125.9536	2.218327	10	121.0109	130.8963
5	10	Non-soluble	n	132.4677	2.218327	10	127.5249	137.4104
6	15	Non-soluble	n	130.5857	2.218327	10	125.6429	135.5284
16	0	control	y	137.3502	2.218327	10	132.4074	142.2929

showing the simple effects of dosage at the given drug types

The same results are obtained when we find the simple effect of drug type at given dosage levels. At dosage level of 10 units, the simple effect for soluble and non-soluble drugs are 127 and 132 respectively. While at a dosage level of 15 units, the simple effects of soluble and non-soluble drugs are 126 and 131 respectively.

```
# Estimated marginal means for drug at each dosage level correct model
bp.m1.emmeans3 <- emmeans (bp.m1, ~ Drug | Dosage)

# Print as formatted table
kable(
  bp.m1.emmeans3,
  caption = "Simple effects of drug types at the given dosage levels"
)
```

Table 8: Simple effects of drug types at the given dosage levels

	Drug	Dosage	Control	emmean	SE	df	lower.CL	upper.CL
4	Soluble	10	n	127.3826	2.218327	10	122.4398	132.3253
5	Non-soluble	10	n	132.4677	2.218327	10	127.5249	137.4104
7	Soluble	15	n	125.9536	2.218327	10	121.0109	130.8963
8	Non-soluble	15	n	130.5857	2.218327	10	125.6429	135.5284
12	control	0	y	137.3502	2.218327	10	132.4074	142.2929

showing the simple effects of drug types at the given dosage levels

Recall, the 2×2 factorial design with control enables us to also compare the main effect of treatments to the control group. This is possible because of the variable Control in the last column of the data set. It serves as an indicator variable and thus enables the model to average over the values of the treatments and compare them to the control group. From Table 9, we see that on average the estimated blood pressure for individuals receiving a treatment compared to control group was 129 and 137 respectively. Thus, the blood pressure for individuals in the control group is higher on average compared to individuals who receive some type of treatment. The difference between the two values is shown by the contrast statement. From Table 9, we see that the difference is 8.25 and with a standard deviation of 2.48. The p-value is 0.01, showing that there is a statistically significant difference between individuals that received some sort of treatment and the control group at $\alpha = 0.05$.

```
# Compare main effect of treatment with the control
bp.m1.emmeans4 <- emmeans (bp.m1, ~ Control)
```

```
summary(bp.m1.emmeans4)
```

	Control	emmean	SE	df	lower.CL	upper.CL
n		129	1.11	10	127	132
y		137	2.22	10	132	142

Results are averaged over the levels of: Drug, Dosage
Confidence level used: 0.95

```
# Difference between treatment and control
contrast(bp.m1.emmeans4, method = "trt.vs.ctrl")
```

	contrast	estimate	SE	df	t.ratio	p.value
y - n		8.25	2.48	10	3.328	0.0076

Results are averaged over the levels of: Drug, Dosage

Finally, Table 10 shows differences between pairs of treatment. This is obtained by writing contrast statements. Writing contrasts is very useful especially when one has so many treatments and may only be interested in a few comparisons. Therefore, we would write some contrast to determine if there is a significant difference between soluble and non-soluble drugs, soluble and control, non-soluble and control. We used the dunnett adjustment to control for family-wise error.

From Table 10, there is no significant difference in effect between soluble and non-soluble drug and between non-soluble drug and control, p-values are 0.13 and 0.14 respectively at 5 percent significance level. Nevertheless, there is a significant difference in effect between soluble drug and the control group, p-value = 0.01.

```
# Compare treatments excluding control
contrast(bp.m1.emmeans, method = "pairwise", adjust = "dunnett")
```

	contrast	estimate	SE	df	t.ratio	p.value
Soluble n - (Non-soluble n)		-4.86	2.22	10	-2.190	0.1314
Soluble n - control y		-10.68	2.72	10	-3.932	0.0076
(Non-soluble n) - control y		-5.82	2.72	10	-2.143	0.1414

Results are averaged over the levels of: Dosage
P value adjustment: dunnettx method for 3 tests

Conclusion

In this chapter, we considered the 2×2 factorial design with control and blocks, and how experiments of this nature can be analyzed. We first considered the case without block for easy understanding and extended the concepts to the case where blocks are considered. The example used to illustrate the concepts in this chapter was about evaluating the effects of two drugs on blood pressure which consisted of two factors and a control group. This design allowed researchers to answer some important questions which are usually present in studies that may be groundbreaking or there may not be enough research already done in that field.

The analysis showed that, without the application of any treatment, blood pressure among individuals may stay high in comparison to when a treatment is administered. Moreover, the type of the drug, whether soluble or not may have an effect on blood pressure. Nevertheless, there was no statistical evidence to support the effect caused by an increase in dosage. Also, for the example and data set used for analysis, there is no interaction between the two factors. Though in this illustration we simulated our data, the concepts are applicable to several studies of this nature. We also want to highlight how analyzing experiments of this nature as a traditional factorial design can change the inference space of the study.